

Advanced Sorting Web Application - 12-Factor App Adherence Report

1. Codebase

- Description: The application must have a single codebase tracked in a version control system, with multiple environments (e.g., development, production).
- Application Status:
 - The project is hosted on [GitHub](https://github.com/newtronahmed/AdvancedSortingWebApplication), indicating a single codebase for development and deployment.
 - Adherence Level:  Fully adheres (One codebase tracked in Git).

2. Dependencies

- Description: All dependencies should be explicitly declared.
- Application Status:
 - The project uses `pom.xml` (Maven) to declare dependencies like `Spring Boot`, `Thymeleaf`, and other libraries.
 - Adherence Level:  Fully adheres (Dependencies declared via Maven).
 - Example: `pom.xml` contains:

```
``xml
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
```

...

3. Config

- Description: Configuration should be stored in the environment, not in the code.
- Application Status:
 - Configuration for database connection, server port, and other settings is stored in `application.properties` or `application.yml`, which can be externalized using environment variables.
 - Adherence Level:  Partially adheres (Configurations can be externalized but currently stored in source code).
 - Recommendation: Move sensitive configs like DB credentials to environment variables.

4. Backing Services

- Description: Treat backing services such as databases and queues as attached resources.
- Application Status:
 - The application uses an in-memory H2 database during development. A switch to a production database like MySQL or PostgreSQL can be made easily by changing the connection string.
 - Adherence Level:  Fully adheres (Database can be swapped with minimal changes).
 - Example:

```
``properties
spring.datasource.url=jdbc:h2:mem:testdb
spring.datasource.url=jdbc:mysql://localhost:3306/prod
``
```

5. Build, Release, Run

- Description: Strict separation between build, release, and run stages.
- Application Status:
 - The application uses Maven for the build process and `Dockerfile` for packaging the application.
 - Build stage: `mvn package`
 - Release: Docker image is built and tagged.
 - Run: Container is launched with environment-specific configurations.
- Adherence Level:  Fully adheres (Build and run stages are distinct).
- Example:

```
``bash
mvn clean package
docker build -t sorting-app .
``
```

6. Processes

- Description: Execute the app as stateless processes.
- Application Status:
 - The web application is stateless, meaning no data is stored in local memory, and each request is independent.
 - Adherence Level:  Fully adheres (Stateless web processes).
 - Recommendation: If user sessions are needed, use Redis or another external session store.

7. Port Binding

- Description: Export services via port binding.
- Application Status:
 - The Spring Boot application binds to a specified port (e.g., 8080) and exposes the service.
 - Adherence Level:  Fully adheres (Port binding via embedded Tomcat).
 - Example:

```
``properties
server.port=8080
``
```

8. Concurrency

- Description: Scale out via the process model.

- Application Status:

- The application can be scaled horizontally by running multiple instances in containers or VMs.

- Adherence Level:  Fully adheres (Scalable via containerization).

- Recommendation: Deploy using Docker or Kubernetes for true scalability.

9. Disposability

- Description: Maximize robustness with fast startup and graceful shutdown.

- Application Status:

- The application uses Spring Boot, which has fast startup and shutdown capabilities.

- Adherence Level:  Fully adheres (Fast startup and shutdown).

- Example:

- Spring Boot logs showing startup time:

...

Started SortingApplication in 4.324 seconds

...

10. Dev/Prod Parity

- Description: Keep development, staging, and production as similar as possible.

- Application Status:

- The application can be easily moved from development (H2 database) to production (MySQL), with minimal changes.

- Adherence Level:  Partially adheres (Environments are close but not identical).

- Recommendation: Set up Docker Compose to align dev and prod environments more closely.

11. Logs

- Description: Treat logs as event streams.

- Application Status:

- The application outputs logs via Spring Boot's logging system, which can be piped to external log aggregators like ELK or Datadog.

- Adherence Level:  Fully adheres (Logs handled as event streams).

- Example:

- Logs streamed to console:

- ...

- 2024-09-22 14:12:34 INFO SortingApplication - Application started

- ...

12. Admin Processes

- Description: Run admin/management tasks as one-off processes.

- Application Status:

- Administrative tasks like database migrations can be run using tools such as Flyway or Liquibase.

- Adherence Level:  Fully adheres (One-off processes can be easily run).

- Recommendation: Implement Flyway or Liquibase for database migrations.

Overall Compliance Summary

- Level of Adherence:

- The Advanced Sorting Web Application largely follows the 12-factor app principles, with strong adherence in areas like dependencies, stateless processes, and port binding.

- Areas for improvement include config management (externalizing configurations), dev/prod parity, and admin process automation.

- Recommendations:

- Move sensitive configuration (like DB credentials) to environment variables.

- Align development and production environments more closely, potentially using Docker Compose.

- Implement automated migrations using Flyway or Liquibase.

Conclusion

This report outlines how the Advanced Sorting Web Application aligns with the 12-factor app methodology, ensuring it is scalable, maintainable, and deployable across different environments. By addressing the highlighted areas for improvement, the application can achieve full adherence to these best practices.

This format is comprehensive and clearly presents the current adherence of your project to the 12-factor principles, along with specific examples and recommendations for improvement. It can be used as either a written report or presented in slides.